Atomic direct limits for interval-preserving maps

A partial answer to an open question in arXiv:2208.13813

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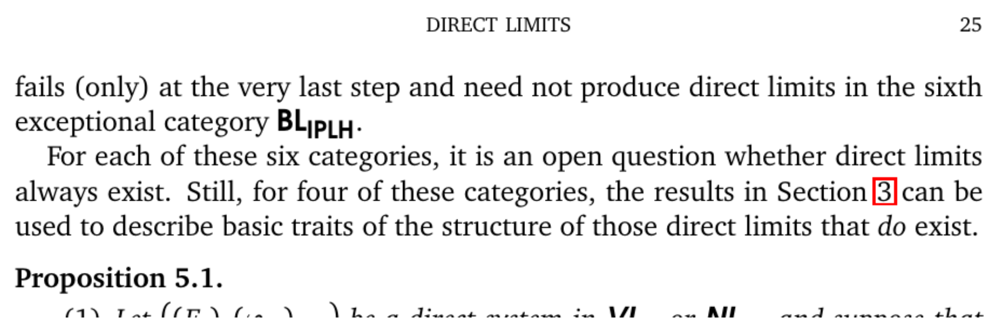
Status. Candidate substantial partial result, likely valid, pending human review. The general existence question remains open, but every direct system in $\mathsf{VL}_{\mathsf{IP}}$ whose objects are lattice-isomorphic to spaces $c_{00}(A)$ has a direct limit in $\mathsf{VL}_{\mathsf{IP}}$. This includes systems of finite-dimensional *Archimedean* vector lattices. The qualifier is necessary because the source allows non-Archimedean vector lattices.

1 Source question

Ding and de Jeu study direct limits in categories of vector lattices, normed lattices, and Banach lattices [1]. On source PDF page 25 they state that it is open whether direct limits always exist in each of six exceptional categories. The first is $\mathsf{VL}_{\mathsf{IP}}$: real vector lattices with interval-preserving linear maps. The other five impose norm or completeness conditions. The theorem below solves a broad algebraically atomic subcase of the first question.



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fails (only) at the very last step and need not produce direct limits in the sixth exceptional category $\mathbf{BL}_{\mathbf{IPLH}}$.

For each of these six categories, it is an open question whether direct limits always exist. Still, for four of these categories, the results in Section 3 can be used to describe basic traits of the structure of those direct limits that *do* exist.

Proposition 5.1.

A positive linear map $T : E \to F$ is interval preserving when

$$T([0,x]_E) = [0,Tx]_F \qquad (x \in E_+).$$

An atom is a nonzero $a \geq 0$ for which $[0,a]$ is contained in the one-dimensional ray $\mathbb{R}a$. For a set A, $c_{00}(A)$ denotes the vector lattice of finitely supported real functions on A, with coordinatewise order.

2 Proof intuition

An interval-preserving map out of $c_{00}(A)$ has an elementary form: every coordinate atom is sent either to zero or to a positive multiple of an atom. Different coordinates may merge onto the same atomic ray. Thus a direct system is combinatorially a directed system of weighted partial maps between sets of atoms. In its ordinary vector-space direct limit, the surviving atomic rays form a basis. Giving that basis the coordinatewise order produces the desired vector lattice. The same atomic description forces every factorization map in the universal property to be interval preserving.

3 Atomic characterization

Lemma 1. *Let A be a set, let F be a vector lattice, and let $T : c_{00}(A) \to F$ be linear. Then T is interval preserving if and only if for every coordinate vector e_a, the vector Te_a is either zero or an atom of F.*

Proof. If T is interval preserving, it is positive and

$$[0, Te_a] = T([0, e_a]) = \{tTe_a : 0 \le t \le 1\}.$$

Consequently a nonzero Te_a is an atom.

Conversely, suppose that every Te_a is zero or an atom. Then T is positive. Fix $x = \sum_{a \in S} x_a e_a \ge 0$, where S is finite, and let $0 \le y \le Tx$. Group the nonzero vectors Te_a for $a \in S$ according to their positive rays. Choose one atom b_r on every resulting ray and write

$$Tx = \sum_r s_r b_r,$$

where $s_r > 0$ and the b_r are pairwise disjoint. The Riesz decomposition property gives $y = \sum_r y_r$ with $0 \le y_r \le s_r b_r$. Atomicity gives $y_r = t_r b_r$ for some $0 \le t_r \le s_r$.

Within a fixed group write $Te_a = \beta_a b_r$. Since $s_r = \sum_a x_a \beta_a$, one can choose numbers $0 \le z_a \le x_a$ with $\sum_a z_a \beta_a = t_r$; this is just the fact that the sum of the intervals $[0, x_a \beta_a]$ is $[0, s_r]$. Doing this for every group and putting $z = \sum_{a \in S} z_a e_a$ gives $0 \le z \le x$ and $Tz = y$. Hence $[0, Tx] \subseteq T([0, x])$; the reverse inclusion follows from positivity. □

4 Existence of the direct limit

Theorem 2. *Let (E_i, ϕ_{ji}) be a direct system over an arbitrary directed set in $\mathsf{VL_{IP}}$. Suppose that every E_i is lattice-isomorphic to $c_{00}(A_i)$ for some set A_i. Then the system has a direct limit in $\mathsf{VL_{IP}}$, and its underlying vector lattice is lattice-isomorphic to $c_{00}(C)$ for a set C of surviving atomic rays.*

Proof. Choose atomic coordinate vectors $e_{i,a}$ in every E_i. By Lemma 1, each connecting map sends $e_{i,a}$ either to zero or to a positive multiple of a coordinate atom at the later stage.

Let (V, q_i) be the ordinary algebraic direct limit of the underlying vector spaces. Consider all nonzero vectors $q_i(e_{i,a})$ and identify two of them when they lie on the same positive ray in V. Let C be the set of the resulting rays, and choose a positive generator u_c from every ray.

We claim that $(u_c)_{c \in C}$ is a vector-space basis of V. It spans: every element of V is the image of an element at one stage, and every such element has finite atomic support. For independence, take a finite relation among distinct u_c and represent all of its terms by coordinate atoms at their respective stages. Directedness provides one later stage k receiving all of them. No representative can map to zero there, since its image in V is nonzero. Each therefore maps to a positive multiple of a coordinate atom of E_k. Two representatives cannot land on the same coordinate atom, since then their images in V would lie on the same positive ray. Their images at stage k are consequently distinct coordinate vectors and are linearly independent. The original relation is trivial.

Use this basis to order V coordinatewise:

$$V_+ = \left\{ \sum_{c \in C} \alpha_c u_c : \alpha_c \ge 0 \text{ and only finitely many } \alpha_c \ne 0 \right\}.$$

Then V is a vector lattice isomorphic to $c_{00}(C)$. Each q_i sends a coordinate atom to zero or to a positive multiple of a coordinate atom of V, so Lemma 1 says that every q_i is interval preserving.

It remains to check the universal property in the smaller category, not only among vector spaces. Let $T_i : E_i \to F$ be any compatible cocone of interval-preserving maps into a vector lattice F. The vector-space direct limit supplies a unique linear map $T : V \to F$ with $Tq_i = T_i$. Every basis atom u_c is a positive multiple of some $q_i(e_{i,a})$, so Tu_c is a positive multiple of $T_i(e_{i,a})$. Lemma 1 shows that Tu_c is zero or an atom of F. Applying that lemma once more proves that T is interval preserving. Thus (V, q_i) has precisely the universal property of a direct limit in $\mathsf{VL}_{\mathsf{IP}}$. $\square$

Corollary 3. *Every direct system of finite-dimensional Archimedean vector lattices and interval-preserving linear maps has a direct limit in* $\mathsf{VL}_{\mathsf{IP}}$.

Proof. Every finite-dimensional Archimedean vector lattice is lattice-isomorphic to $\mathbb{R}^n = c_{00}(\{1, \ldots, n\})$ with coordinatewise order, so Theorem 2 applies. $\square$

5 Upgrade attempts, novelty, and limits

The first proof attempt covered only coordinatewise finite-dimensional objects. The key observation that every element uses finitely many atoms upgrades the argument to arbitrary $c_{00}(A_i)$ and arbitrary directed index sets. A further attempt to remove Archimedeanity from the finite-dimensional corollary does not follow from this method: finite-dimensional non-Archimedean lattices, such as lexicographically ordered spaces, need not be spanned by atoms. Likewise $\ell^\infty(A)$ is atomic in the order-theoretic sense but is not the algebraic span of its atoms, so the basis construction does not cover it.

The result also does not automatically extend to the five normed or Banach categories in the source question. Quotient norms and completion introduce conditions absent from the algebraic colimit, and completion can add elements that are not finite sums of atoms; interval preservation need not survive the last inclusion into the completion.

A bounded novelty search covered the run-wide result and attempt indexes, the arXiv identifier, exact phrases around the source question, and combinations of "atomic vector lattice", "interval preserving", and "direct limit". No primary source stating this atomic subcase was found. Because the proof is elementary and categorical, novelty confidence is moderate; correctness confidence is high. Human review should focus on the basis-independence argument and the repeated use of Lemma 1 in the universal property.

References

[1] C. Ding and M. de Jeu, *Direct limits in categories of normed vector lattices and Banach lattices*, arXiv:2208.13813.